

1.2.6. Variational method

For a particle under an external force

$$W(\vec{r}t | \vec{r}_0 t_0) = \int_{C[\vec{r}_0 t_0; \vec{r}t]} \prod_{t_0}^t \frac{d\vec{r}(t)}{\sqrt{4D}} \exp \left\{ -\frac{1}{4D} \int_{t_0}^t \left(\frac{d\vec{r}}{d\tau} \right)^2 d\tau + \frac{1}{2\gamma D} \int_{t_0}^t d\vec{r} \cdot \vec{F} - \int_{t_0}^t d\tau V(\vec{r}(t)) \right\}$$

The exp. term

$$\Omega = \exp \left\{ -\frac{1}{4D} \int_{t_0}^t d\tau L(\vec{r}(t)) \right\} \quad (1.2.106)$$

with (use (1.2.92) for $V(r)$):

$$\begin{aligned} L &= \left(\frac{d\vec{r}}{d\tau} \right)^2 + 4DV(\vec{r}(t)) - \frac{2}{\gamma} \vec{F} \cdot \frac{d\vec{r}}{d\tau} \\ &= \left(\frac{d\vec{r}}{d\tau} \right)^2 + 2D \left[\frac{F^2}{4\gamma^2 D} + \frac{\nabla \cdot \vec{F}}{2\gamma} \right] - \frac{2}{\gamma} \vec{F} \cdot \frac{d\vec{r}}{d\tau} \\ &= \left(\frac{d\vec{r}}{d\tau} \right)^2 - \frac{2}{\gamma} \vec{F} \cdot \frac{d\vec{r}}{d\tau} + \left(\frac{F}{\gamma} \right)^2 + \frac{2D}{\gamma} \text{div} \vec{F} \\ &= \left(\frac{d\vec{r}}{d\tau} - \frac{\vec{F}}{\gamma} \right)^2 + \frac{2D}{\gamma} \text{div} \vec{F} \end{aligned} \quad (1.2.107)$$

• When $D \rightarrow 0$, fluctuations are switched off, the particle goes along the ~~classical~~ ^{deterministic} fixed trajectory; $\exp(\dots)$ in W should be max, and if $D \rightarrow 0$ then L should be zero, which from (1.2.107) at $D \rightarrow 0$ gives

$$\int_{t_0}^t \left(\frac{d\vec{r}}{d\tau} - \frac{\vec{F}}{\gamma} \right)^2 d\tau = 0 \rightarrow \frac{d\vec{r}}{d\tau} = \frac{\vec{F}}{\gamma}$$

which is the result at $m=0$ (or very large friction in the medium)

• For any D all trajectories contribute to Ω , but those which minimise the integral - most;

$$S = \int_{t_0}^t d\tau L(\vec{F}(\tau))$$

$$\delta S = \delta \int_{t_0}^t d\tau L(\vec{F}(\tau)) = 0 \quad (1.2.110)$$

giving: $\frac{\partial L}{\partial x} = \frac{\partial}{\partial \tau} \frac{\partial L}{\partial \dot{x}}, (y, z) \quad (1.2.111)$

• Hence, ~~and~~ for small $\mathcal{Q}$ an approximate expression for W ,

$$W(\vec{F}|t_0) \simeq \phi(t-t_0) \exp \left\{ -\frac{1}{4\mathcal{Q}} \int_{t_0}^t d\tau L(\vec{F}_c(\tau)) \right\} \quad (1.2.112)$$

$\vec{F}_c(t)$ - trajectory from (1.2.111) - "classical" trajectory,
 $\phi(t-t_0)$ - only depends on time (the ~~fluctuation~~ "fluctuation factor")
 $\hookrightarrow$ for quadratic functions $L(\vec{F}(t))$.

L - is not actual Lagrangian

Example 1.6

$$W(x|t/x_0, 0) = \int_{C[x_0, 0; x, t]} d_w x(\tau) = \int_{C[...]} \prod \frac{dx(\tau)}{\sqrt{\pi d\tau}} \exp \left[\frac{i}{2} \int_0^t \dot{x}^2 d\tau \right]$$

Here $L = \frac{1}{4\mathcal{Q}} \left(\frac{dx}{dt} \right)^2 = \left(\frac{dx}{dt} \right)^2 = \dot{x}^2 \quad (\mathcal{Q} = 1 \text{ here})$

Euler eqs: $\frac{\partial}{\partial \tau} \frac{\partial L}{\partial \dot{x}} = 0 \rightarrow \frac{\partial}{\partial \tau} (2\dot{x}) = 2\ddot{x} = 0$

$\hookrightarrow x_c(t) = At + B \quad ; \quad x_c(0) = x_0, x_c(t) = x \quad (1.2.117)$

$\hookrightarrow x_c(t) = x_0 + \frac{t}{T} (x - x_0) \quad (1.2.118)$

Therefore,

$$\int_0^t L d\tau = \int_0^t \dot{X}_c^2 d\tau = \int_0^t \left(\frac{(x-x_0)}{t} \right)^2 d\tau = \left(\frac{x-x_0}{t} \right)^2 t = \frac{(x-x_0)^2}{t}$$

and

$$W = \phi(t) \exp\left(-\frac{(x-x_0)^2}{t}\right) \quad (1.2.113)$$

$\phi(t)$ can be obtained from normalisation:

$$\int_{-\infty}^{\infty} W dx = 1 \rightarrow \phi(t) = \frac{1}{\sqrt{\pi t}} \rightarrow W(x|t|x_0) = \frac{1}{\sqrt{\pi t}} e^{-\frac{(x-x_0)^2}{t}}$$

which is exact in this case.

(1.1.119)

Now use $X_c(t)$ as the mean trajectory: $X(t) = X_c(t) + X(t)$
 $X(0) = X_0 = X_c(0)$
 $X(t) = X = X_c(t)$

so that $X(0) = X(t) = 0$

(1.1.120)

The Jacobian = 1:

$$W(x|t|x_0) = \exp\left(-\int_0^t \dot{X}_c^2 d\tau\right) \left[\int_{C[0,0;t]} \frac{dX(t)}{\sqrt{\pi t}} \exp\left(-\int_0^t \dot{X}^2 d\tau\right) \right] \quad (1.1.121)$$

$X(t)$ - deviation from the "classical" trajectory.

$$\text{Here } \dot{X}^2 = (\dot{X}_c + \dot{X})^2 = \dot{X}_c^2 + 2\dot{X}_c\dot{X} + \dot{X}^2,$$

$$\text{But } \int_0^t 2\dot{X}_c\dot{X} d\tau = 2 \frac{x-x_0}{t} \int_0^t \dot{X} d\tau = 2 \frac{x-x_0}{t} \underbrace{[X(t) - X(0)]}_{=0} = 0$$

That is why the cross term does not contribute in (1.1.121).

Therefore,

$$\phi(t) = \int_{C[0,0;t]} dX(t) \equiv \frac{1}{\sqrt{\pi t}} \quad \left(\begin{array}{l} \text{see example 1.1} \\ \text{discretisation?} \\ \text{method or simply} \\ \text{as } W(0|0) \end{array} \right)$$

• Example 1.7

$$I_7 = \int_C [x_0, 0; x, t] d_w x(\tau) \exp \left[-k^2 \int_0^t x^2 d\tau \right] \quad (1.2.126)$$

Here

$$L = \left(\frac{1}{42} \right) \ddot{x}^2 + k^2 x^2 = \ddot{x}^2 + k^2 x^2$$

1 (units)
of time

$$\frac{\partial L}{\partial x} = \frac{\partial}{\partial \tau} \frac{\partial L}{\partial \dot{x}} \rightarrow 2k^2 x = 2\ddot{x} \rightarrow \ddot{x} = k^2 x \quad \text{for } x_c(\tau)$$

$$x_c(\tau) = C_1 e^{k\tau} + C_2 e^{-k\tau}$$

$$x_c(0) = x_0, x_c(t) = x \rightarrow \begin{cases} C_1 + C_2 = x_0 \\ C_1 e^{kt} + C_2 e^{-kt} = x \end{cases}$$

~~$$C_1 (e^{kt} - e^{-kt}) = x \rightarrow C_1 2 \sinh(kt) = x, C_2 = \frac{x}{2 \sinh(kt)}$$~~

$$C_1 = \frac{\begin{vmatrix} x_0 & 1 \\ x & e^{-kt} \end{vmatrix}}{\begin{vmatrix} 1 & 1 \\ e^{kt} & e^{-kt} \end{vmatrix}} = \frac{x_0 e^{-kt} - x}{e^{-kt} - e^{kt}} = \frac{1}{2} \frac{x - x_0 e^{-kt}}{\sinh(kt)}$$

$$C_2 = \frac{\begin{vmatrix} 1 & x_0 \\ e^{kt} & x \end{vmatrix}}{-2 \sinh(kt)} = -\frac{x - x_0 e^{kt}}{2 \sinh(kt)}$$

$$\dot{x} = \dot{x}_c + \dot{x} = (C_1 e^{k\tau} - C_2 e^{-k\tau})k + \dot{x}$$

$$\int_0^t \dot{x}^2 d\tau = \int_0^t \dot{x}_c^2 d\tau + \int_0^t \dot{x}^2 d\tau + 2 \int_0^t \dot{x}_c \dot{x} d\tau$$

~~$$2k \int_0^t k^2 x^2 d\tau = k^2 \int_0^t (x_c + x)^2 d\tau = k^2 \int_0^t x_c^2 d\tau + k^2 \int_0^t x^2 d\tau +$$~~

$$+ 2k^2 \int_0^t x_c x d\tau$$

and

$$\int_0^t (\dot{x}^2 + k^2 x^2) d\tau = \int_0^t (\dot{x}_c^2 + k^2 x_c^2) d\tau + \int_0^t (\dot{x}^2 + k^2 x^2) d\tau +$$

$$+ 2 \int_0^t (\dot{x}_c \ddot{x}_c + k^2 x_c \dot{x}_c) d\tau$$

In the last integral we take the 1st ~~part~~ ^{term} by parts:

$$\int_0^t \dot{x}_c \ddot{x}_c d\tau = \underbrace{\dot{x}_c x_c} \Big|_0^t - \int_0^t x_c \ddot{x}_c d\tau = - \int_0^t x_c k^2 x_c d\tau$$

and it cancels out with the 2nd term; hence:

$$I_7 = e^{-\int_0^t (\dot{x}_c^2 + k^2 x_c^2) d\tau} \underbrace{\left[\int_{C[00;0t]} d_w X(\tau) \exp \left[- \int_0^t k^2 X^2 d\tau \right] \right]}_{\phi(t)}$$

where

$$\begin{aligned} \int_0^t (\dot{x}_c^2 + k^2 x_c^2) d\tau &= \int_0^t d\tau \left\{ \left(\dot{c}_1 k e^{k\tau} + k c_2 e^{-k\tau} \right)^2 + \right. \\ &+ k^2 \left(c_1 e^{k\tau} + c_2 e^{-k\tau} \right)^2 \Big\} = \int_0^t d\tau \left\{ c_1^2 k^2 e^{2k\tau} + k^2 c_2^2 e^{-2k\tau} - \right. \\ &\quad \left. \cancel{2k^2 c_1 c_2} + k^2 c_1^2 e^{2k\tau} + k^2 c_2^2 e^{-2k\tau} + \cancel{2k^2 c_1 c_2} \right\} \\ &= k^2 \int_0^t d\tau \left\{ 2c_1^2 e^{2k\tau} + 2c_2^2 e^{-2k\tau} \right\} \\ &= k^2 \left\{ 2c_1^2 \frac{e^{2kt} - 1}{2k} + 2c_2^2 \frac{e^{-2kt} - 1}{-2k} \right\} \\ &= \frac{k}{2} \left\{ c_1^2 (e^{2kt} - 1) + c_2^2 (e^{-2kt} - 1) \right\} \\ &= k \left\{ (e^{2kt} - 1) \frac{(x - x_0 e^{-kt})^2}{4 \sinh^2(kt)} + (1 - e^{-2kt}) \frac{(x - x_0 e^{kt})^2}{4 \sinh^2(kt)} \right\} \end{aligned}$$

$$\begin{aligned}
 &= \frac{k}{4 \sinh^2(kt)} \left\{ x^2 \left[\cancel{e^{2kt}} - \cancel{1} + \cancel{1} - e^{-2kt} \right] - \right. \\
 &- 2xx_0 \left[(e^{2kt} - 1)e^{-kt} + (1 - e^{-2kt})e^{kt} \right] \\
 &+ x_0^2 \left[e^{-2kt}(e^{2kt} - 1) + e^{2kt}(1 - e^{-2kt}) \right] \left. \right\} \\
 &= \frac{k}{4 \sinh^2(kt)} \left\{ x^2 \cdot 2 \sinh(2kt) - 2xx_0 \left[\underbrace{e^{kt} - e^{-kt} + e^{kt} - e^{-kt}}_{2(e^{kt} - e^{-kt}) = 4 \sinh(kt)} \right] \right. \\
 &+ x_0^2 \left[\underbrace{\cancel{e^{2kt}} - \cancel{e^{-2kt}} + e^{2kt} - 1}_{2 \sinh(2kt)} \right] \left. \right\}
 \end{aligned}$$

As $\sinh(2k) = 2 \sinh(kt) \cosh(kt)$,

$$= \frac{k}{\sinh(kt)} \left\{ x^2 \cosh(kt) - 2xx_0 + x_0^2 \cosh(kt) \right\},$$

we

$$\boxed{I_7 = \phi(t) \exp \left\{ -\frac{k}{\sinh(kt)} \left[(x^2 + x_0^2) \cosh(kt) - 2xx_0 \right] \right\}} \quad (1.2.135)$$

Normalisation gives:

$$\phi(t) = \int_{\mathcal{C}[00;0t]} d_w x(\tau) \exp \left[-\int_0^t k^2 x^2 d\tau \right] = \sqrt{\frac{k}{\pi \sinh(kt)}} \quad (1.2.136)$$

We also did this directly using discretisation from the general formula containing functional

$$\exp \left[-\int_0^t p(s) x^2(s) ds \right]$$

Example 1.8

$$I_8 = \int_{C[x_0, 0; x, t]} \prod \frac{dx(\tau)}{\sqrt{\pi} d\tau} \exp \left\{ \frac{kt}{2} - \int_0^t d\tau (\dot{x} + kx)^2 \right\}$$

This corresponds (after change of variables) to the Brown. particle in the harmonic field.

Here:

$$\begin{aligned} \int_0^t d\tau (\dot{x} + kx)^2 &= \int_0^t d\tau (\dot{x}^2 + 2kx\dot{x} + k^2x^2) = \int_0^t d\tau \dot{x}^2 + 2k \int_0^t dx^2 + \\ &+ \int_0^t k^2 x^2 d\tau = k(x^2 - x_0^2) + \int_0^t d\tau \dot{x}^2 + k^2 \int_0^t x^2 d\tau \end{aligned}$$

giving

$$I_8 = e^{\frac{kt}{2}} e^{-k(x^2 - x_0^2)} \int_{C[x_0, 0; x, t]} \underbrace{\prod \frac{dx(\tau)}{\sqrt{\pi} d\tau}}_{d_W x(\tau)} \exp \left(- \int_0^t \dot{x}^2 d\tau \right) \exp \left(- k^2 \int_0^t x^2 d\tau \right)$$

The integral is from example 1.7, i.e.

$$I_8 = \exp \left(\frac{kt}{2} - k(x^2 - x_0^2) \right) I_7$$

Example 1.9

B. particle with inertia:

$$W(x, t | x_0, v_0, t_0) = \int_{C[v_0, 0; v, t]} \prod \frac{dv(\tau)}{\sqrt{\pi} d\tau} \exp \left(- \int_0^t \dot{v}^2 d\tau \right) \delta \left(x(t) - x_0 - \int_0^t v(\tau) d\tau \right) \quad (1.2.135)$$

We use variational method with a Lagrangian multiplier to account for the constraint:

$$x(t) = x_0 + \int_0^t v(\tau) d\tau \quad (1.2.136)$$

We minimise then

$$\delta = \int_0^t \dot{v}^2 d\tau + 2 \left(x(t) - x_0 - \int_0^t v(\tau) d\tau \right)$$

with the Euler eq.

~~$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial x_0}$~~ & here

$$\delta = \lambda(x - x_0) + \int_0^t (\ddot{v}^2 - 2v) d\tau$$

$$\frac{\partial L}{\partial v} = \frac{\partial}{\partial \tau} \frac{\partial L}{\partial \dot{v}} \rightarrow \frac{\partial L}{\partial v} = -\lambda = 2\ddot{v}$$

$$\hookrightarrow \frac{d}{d\tau}(\ddot{v}) = \frac{\lambda}{2} \rightarrow \ddot{v} = -\frac{\lambda}{2}\tau + C_1, v = -\frac{\lambda}{4}\tau^2 + C_1\tau + C_2$$

$$\begin{cases} v(0) = v_0 \\ v(t) = v \end{cases} \rightarrow \begin{cases} v_0 = C_2 \\ v = -\frac{\lambda}{4}t^2 + C_1t + C_2 \end{cases} \rightarrow C_1 = \frac{v-v_0}{t} + \frac{\lambda}{4}t$$

or $v_c(\tau) = -\frac{\lambda}{4}\tau^2 + \left(\frac{v-v_0}{t} + \frac{\lambda}{4}t\right)\tau + v_0$

Substitute into the constraint to find λ :

$$x - x_0 = \int_0^t v(\tau) d\tau = -\frac{\lambda}{4} \frac{t^3}{3} + \left(\frac{v-v_0}{t} + \frac{\lambda}{4}t\right) \frac{t^2}{2} + v_0 t$$

$$x - x_0 - v_0 t - \frac{v-v_0}{2}t = \lambda \left[-\frac{t^3}{12} + \frac{t^3}{8} \right] = \lambda \frac{t^3}{24}$$

$$\hookrightarrow \lambda = \frac{24}{t^3} \left[x - x_0 - \frac{v+v_0}{2}t \right]$$

giving

$$C_1 = \frac{v-v_0}{t} + \frac{t}{4} \frac{24}{t^3} \left[x - x_0 - \frac{v+v_0}{2}t \right] = \frac{v-v_0}{t} + \frac{6}{t^2} \left[x - x_0 - \frac{v+v_0}{2}t \right]$$

Hence, $v = v_c + V_t$

$$W = \phi(t) \exp \left[- \int_0^t \ddot{v}_c(\tau) d\tau \right] (\ddot{v}_c + \ddot{V})^2$$

with V being a deviation from v_c :

$$\begin{aligned} \ddot{v}_c &= -\frac{\lambda}{2}\tau + \frac{v-v_0}{t} + \frac{\lambda}{4}t = -\frac{12}{t^3} \left[x - x_0 - \frac{1}{2}(v+v_0)t \right] \tau \\ &+ \frac{v-v_0}{t} + \frac{6}{t^2} \left[x - x_0 - \frac{1}{2}(v+v_0)t \right] \end{aligned}$$

$$\int_0^t \ddot{v}_c^2(\tau) d\tau = \frac{\lambda^2}{4} \cdot \frac{t^3}{3} - 2 \frac{\lambda}{2} \left[\frac{v-v_0}{t} + \frac{\lambda}{4} t \right] \frac{t^2}{2} + \left[\frac{v-v_0}{t} + \frac{\lambda}{4} t \right]^2 t$$

$$= \frac{\lambda^2 t^3}{12} - \frac{\lambda t^2}{2} \left[\frac{v-v_0}{t} + \frac{\lambda}{4} t \right] + t \left[\frac{v-v_0}{t} + \frac{\lambda}{4} t \right]^2$$

There is also a term

$$\int_0^t 2 \ddot{v}_c \dot{V} = \left[2 \ddot{v}_c V \right]_0^t - 2 \int_0^t \ddot{v}_c V = -2 \int_0^t \ddot{v}_c V = 2 \int_0^t \dot{V} d\tau$$

Here

$$x(t) - x(0) = \int_0^t v(\tau) d\tau = \int_0^t v_c d\tau + \int_0^t V(\tau) d\tau$$

$$\int_0^t v_c d\tau = -\frac{\lambda}{4} \frac{t^3}{3} + \left[\frac{v-v_0}{t} + \frac{\lambda}{4} t \right] \frac{t^2}{2} + v_0 t$$

$$= -\frac{\lambda t^3}{12} + \frac{\lambda t^3}{8} + v_0 t + \frac{1}{2} (v-v_0) t = \frac{1}{2} (v+v_0) t + \frac{\lambda t^3}{24}$$

$$= \frac{1}{2} (v+v_0) t + x - x_0 - \frac{1}{2} (v+v_0) t = x - x_0,$$

i.e. $\int_0^t V(\tau) d\tau = 0,$

as the linear in V term is zero.

$$W(xvt|x_0v_0) = \exp\left[-\int_0^t d\tau \ddot{v}_c^2\right] \underbrace{\int_{C[v_0,0;vt]} \prod \frac{dV(\tau)}{\sqrt{\pi} d\tau} \exp\left[-\int_0^t d\tau \dot{V}^2\right]}_{\phi(t) \text{ [normalisation]}}$$

Here

$$\int_0^t \ddot{v}_c^2 d\tau = \frac{\lambda^2 t^3}{12} - \frac{\lambda t^3}{8} - \frac{\lambda t^2}{2} \frac{v-v_0}{t} + \frac{(v-v_0)^2}{t} + 2 \frac{\lambda}{4} t^2 \frac{v-v_0}{t} +$$

$$+ \frac{\lambda^2 t^3}{16} = \frac{\lambda^2 t^3}{48} - \frac{\lambda t^2}{2} \frac{v-v_0}{t} + \frac{(v-v_0)^2}{t} + \frac{\lambda t^2}{2} \frac{v-v_0}{t}$$

$$= \frac{(v-v_0)^2}{t} + \frac{24^2 t^3}{t^6 \cdot 48} \left[x - x_0 - \frac{v+v_0}{2} t \right]^2$$

$$= \frac{(v-v_0)^2}{t} + \frac{12}{t^3} \left[x - x_0 - \frac{1}{2}(v+v_0)t \right]^2$$

and the normalisation (w.r.t. ~~v~~ ϕ gives and x) gives:

$$\int dx \int dv W(xvt | x_0 v_0 0) = 1$$

Integrate w.r.t x first:

$$\int dx \phi(t) \exp \left\{ -\frac{12}{t^3} \left[x - x_0 - \frac{1}{2}(v+v_0)t \right]^2 - \frac{(v-v_0)^2}{t} \right\}$$

$$= \phi(t) \exp \left\{ -\frac{(v-v_0)^2}{t} \right\} \cdot \int_{-\infty}^{\infty} dx' \exp \left(-\frac{12}{t^3} x'^2 \right)$$

$$= \phi(t) e^{-\frac{(v-v_0)^2}{t}} \cdot \sqrt{\frac{\pi}{12/t^3}} = \sqrt{\frac{\pi t^3}{12}} \phi e^{-\frac{(v-v_0)^2}{t}},$$

now w.r.t v :

$$\int_{-\infty}^{\infty} dv \sqrt{\frac{\pi t^3}{12}} \phi e^{-\frac{(v-v_0)^2}{t}} = \phi \sqrt{\frac{\pi t^3}{12}} \cdot \sqrt{\frac{t}{1/t}}$$

$$= \phi \sqrt{\frac{\pi^2 t^4}{12}} = \phi \frac{\pi t^2}{\sqrt{12}}, \quad \text{r.t.}$$

$$\boxed{W = \frac{\sqrt{12}}{\pi t^2} \exp \left\{ -\frac{12}{t^3} \left[x - x_0 - \frac{1}{2}(v+v_0)t \right]^2 - \frac{(v-v_0)^2}{t} \right\}} \quad (1.2.139)$$